

Satellite: A satellite is a man-made object placed in space around the Earth to collect information or for communication. Earth is also technically a satellite because it moves around the Sun.

Satellite Communication: Satellite communication is a system that works like a radio relay station in space above the Earth. It receives amplifies, and sends back analog and digital signals on a specific radio frequency. It is very important in the **global telecommunication** system.

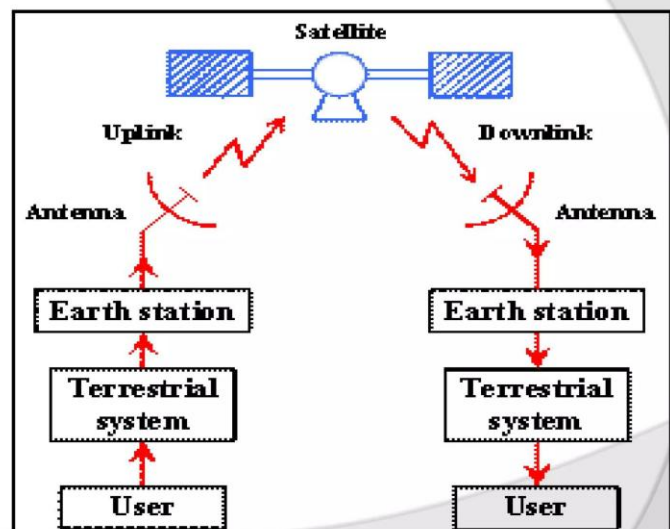
Elements of Satellite Communication:

There are two major elements of a satellite communication system:

1. Space Segment
2. Ground Segment

Basic Flow of Satellite Communication:

a user sends data through a terrestrial(স্থলভ) system → earth station → antenna → uplink to satellite → satellite receives and retransmits → downlink → receiving antenna → earth station → terrestrial system → destination user.



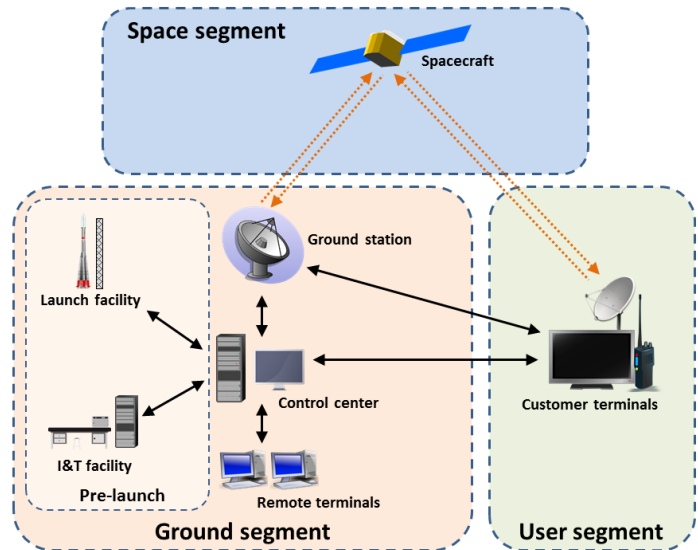
Space Segment includes:

- The satellite itself
- Means for launching the satellite
- Electrical power system
- Mechanical structure
- Communication transponders
- Communication antennas
- Attitude and orbit control system



Ground Segment consists of:

- Earth stations
- Rear ward communication links
- User terminals and interfaces
- Network control center
- Transmit equipment
- Receive equipment
- Antenna system



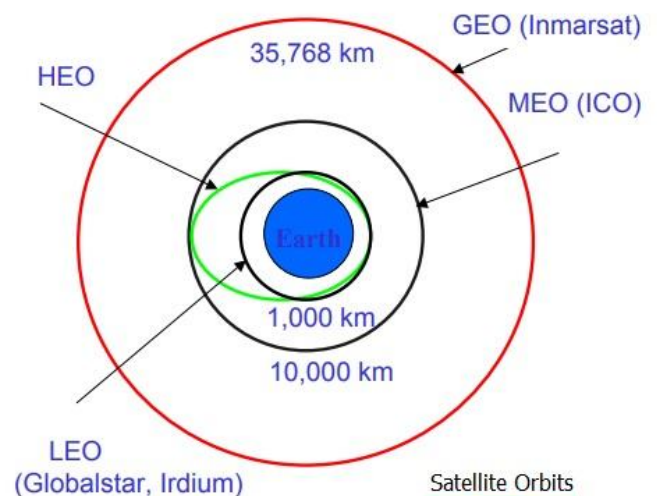
Satellite Control Centre functions:

- Tracking of the satellite
- Receiving data
- Eclipse management of the satellite
- Commanding the satellite for station keeping
- Determining orbital parameters from tracking and ranging data
- Switching ON/OFF of different subsystems as per operational requirements

Orbits for Satellite Communication:

An orbit is the path; a satellite follows around the planet. Satellite orbits are divided into two main categories:

1. Non-Geostationary Orbit (NGSO)
2. Geo Stationary Orbit (GSO)



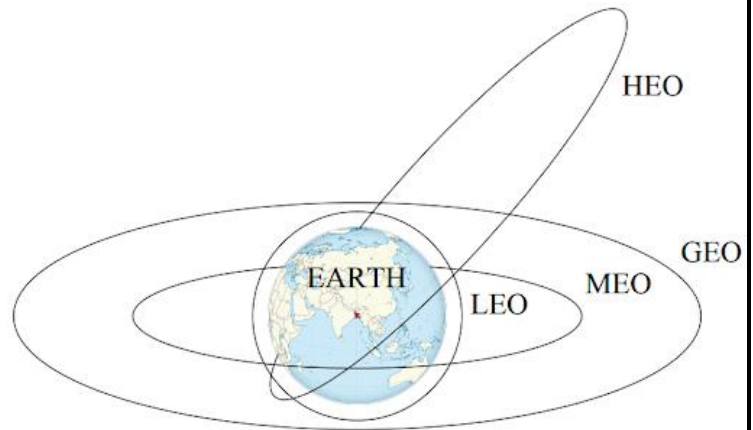
Satellite Orbits by Height:

LEO (Low Earth Orbit):

- Height: 160 to 1600 km above Earth
- Small and easy to launch
- Suitable for mass production
- Used for high-speed data communication

MEO (Medium Earth Orbit)

- Height: 8000 to 18000 km above Earth
- A compromise between LEO and GEO
- Has more delay and needs higher power than LEO



GEO (Geostationary Earth Orbit)

- Height: 35,786 km above the equator
- This is the geostationary orbit

HEO (High Elliptical Orbit)

- Height: 18,500 to 35,000 km above Earth
- Satellite moves close to Earth and then far into space repeatedly
- Gives better coverage to high northern and southern areas



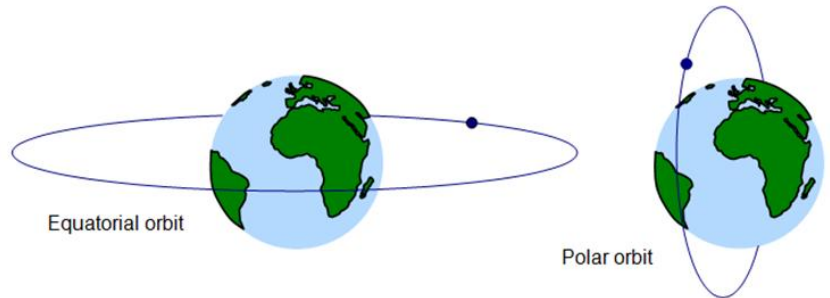
Non-Geostationary Orbit (NGSO):

Early satellite communication used non-geostationary low earth orbits because launch vehicles had technical limitations in placing satellites in higher orbits.

Classification of NGSOs by orbital plane:

Polar Orbit

- The satellite moves from pole to pole
- Inclination angle is exactly 90 degrees



Equatorial Orbit

- The orbital plane moves with the Earth's equatorial plane
- Inclination is zero or very small

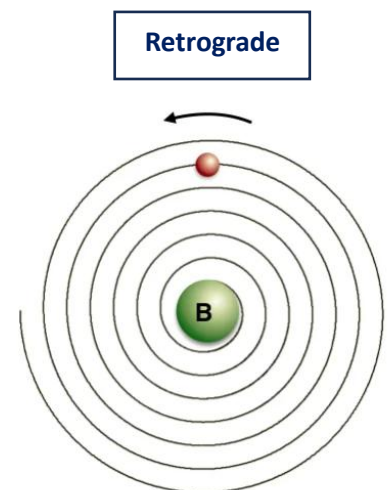
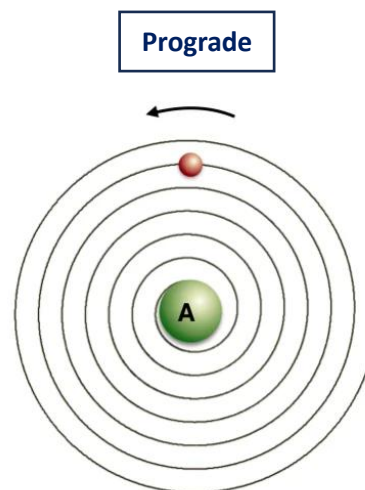
Inclined Orbit

- Any orbit that is neither polar nor equatorial falls into this category

Classification of NGSOs by direction:

Prograde Orbit

- The satellite that moves in the same direction as Earth's rotation
- Inclination is between 0° and 90°



Retrograde Orbit

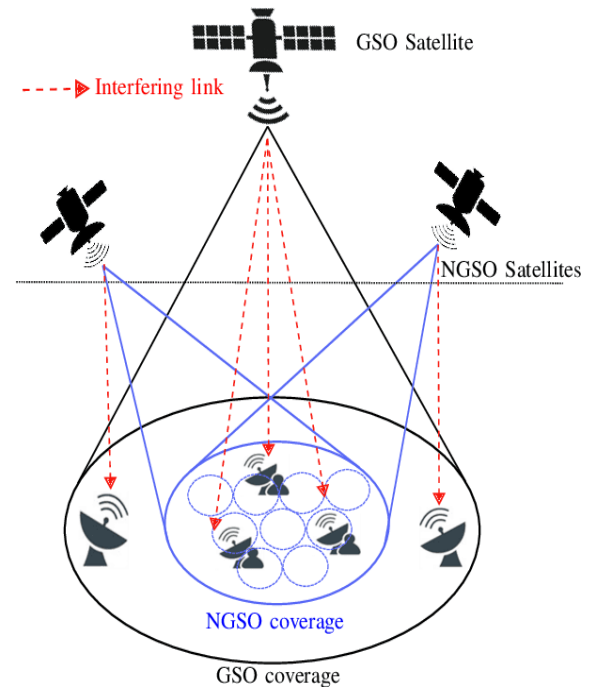
- The satellite moves in the opposite direction to Earth's rotation
- Inclination is between 90° and 180°

Advantages of NGSO:

- Less booster power needed
- Less delay in transmission
- Reduced echo problem in voice communication
- Better for higher latitude areas
- Lower cost to build and launch satellites

Disadvantages of NGSO:

- Complex handoff problem between satellites
- Shorter satellite lifetime
- Requires frequent satellite replacement compared to GSO
- Increases space debris (ধ্বংসাবশেষ) problem
- Needs many satellites for global coverage
- Each LEO satellite covers only a small area for a short time



Geostationary Orbit (GSO):

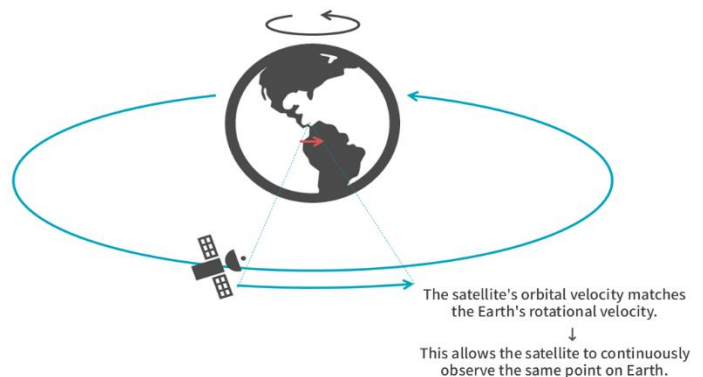
There is only one geostationary orbit around the Earth. It is on the Earth's equatorial plane. A satellite in this orbit moves at the same speed as the Earth rotates, so the satellite appears fixed (stationary) from the ground. As a result,

Because the satellite appears fixed, the ground station antenna does not need to rotate or track it, which reduces cost.

Key Facts about GSO

- One geostationary satellite can cover 38% of the Earth's surface
- Three geostationary satellites can cover the whole Earth
- Doppler shift effect is very small

Geostationary orbit

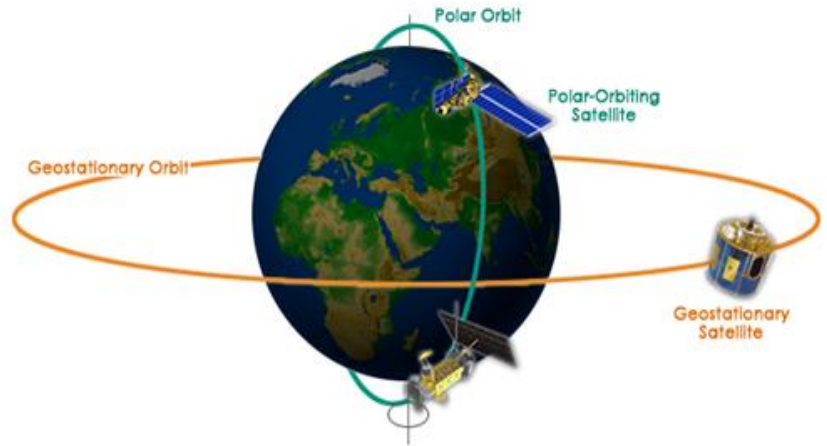


Advantages of GSO

- Simple ground station tracking
- Nearly constant transmission range
- Very small frequency shift
- No need for rotating antenna
- Wide coverage area

Disadvantages of GSO

- Transmission delay is about 250 milliseconds
- Large free space loss because of long distance
- No polar coverage from geostationary orbit



Evolution of Satellite Communication:

During the early 1950s, both passive and active satellites were explored for long-distance communication.

Passive Satellites:

A passive satellite only reflects signals from one earth station to another. It reflects incoming electromagnetic signals without any modification or amplification. It cannot generate power; it only bounces the signal back.

Disadvantages of Passive Satellites

- Earth stations needed very high transmission power
- Large and expensive earth stations with tracking systems were required
- A global system would need many passive satellites used randomly for many users
- Cannot be controlled from the ground
- High signal loss due to long distance between transmitter and receiver

Passive satellites were used in early years, but later they were replaced by active satellites.

Active Satellites:

Active satellites amplify, modify, and retransmit signals that is received from Earth. Any satellite that can transmit power is called an active satellite.

Advantages over Passive Satellites

- Require lower power earth stations
- Less costly to operate
- Not open to random unauthorized use
- Can be controlled from the ground

Disadvantages of Active Satellites

- Require larger and more powerful rockets to launch heavier satellites
- Need on-board power supply
- Service interruptions can occur if electronic components fail

